

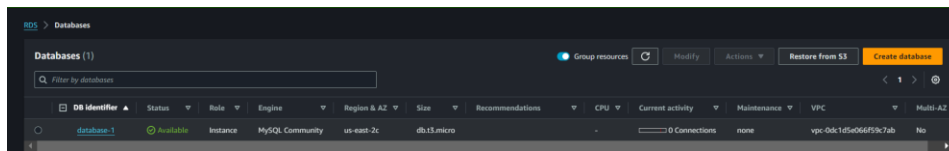
ANGULARSPIRE_SPRING APP PROJECT

Apurva Anil Jagdale

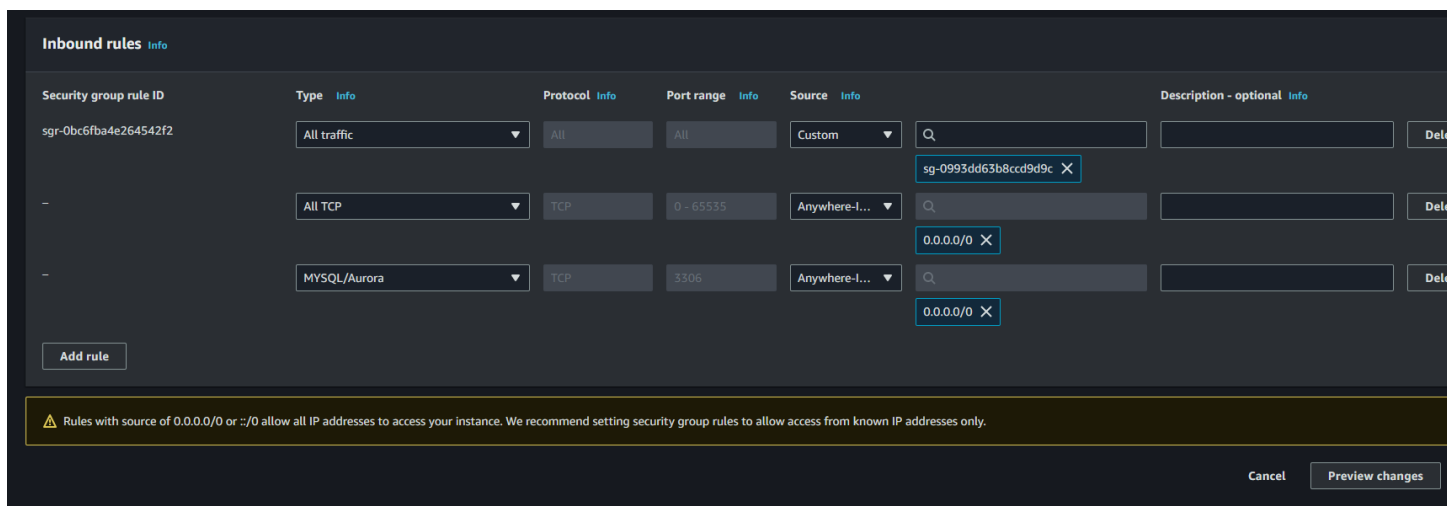
Requirements:

- 1) Create RDS database in AWS.
- 2) Create EC2 Instance for Backend.
- 3) Create EC2 Instance for Frontend.
- 4) Deploy the Application

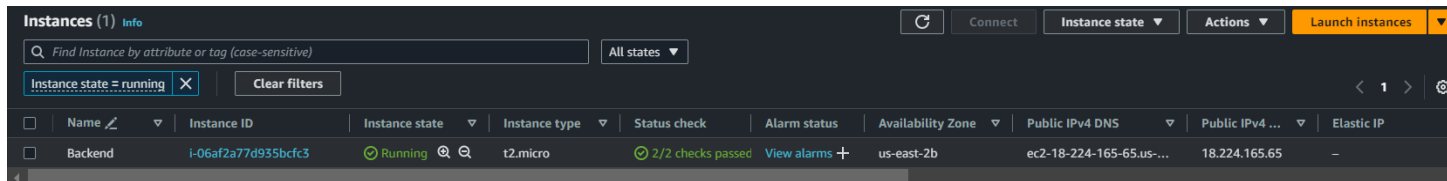
Step I : Create the RDS database in aws.



Step II : In the Database add below security rules.



Step III : Create the EC2 Instance using ubuntu AMI for Backend and connect it.



The screenshot shows the AWS Management Console 'Instances' page. A single instance named 'Backend' is listed with the following details:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
Backend	i-06af2a77d935bfc3	Running	t2.micro	2/2 checks passed	View alarms +	us-east-2b	ec2-18-224-165-65.us-...	18.224.165.65	-

Step IV : give the “sudo apt update -y” command for update the local package index.

```
ubuntu@ip-172-31-26-5:~$ sudo apt update -y
Get:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:2 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
```

Step V : Now,we need the Mariadb server using “sudo apt install mariadb-server -y” command install the mariadb server.

```
ubuntu@ip-172-31-26-5:~$ sudo apt install mariadb-server -y
Reading package lists... Done
```

Step VI : for login into database use below command.

```
ubuntu@ip-172-31-26-5:~$ sudo -i database-1.cjctkwtqfz.us-east-2.elb.amazonaws.com ~$ mysql -padmin123
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MySQL connection id is 27
Server version: 8.0.35 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\q' to clear the current input statement.

mysql>
```

Step VII : create the database use create database <database_name> ; and then exit from db.

```
MySQL [(none)]> CREATE DATABASE springbackend;
Query OK, 1 row affected (0.003 sec)

MySQL [(none)]> exit
Bye
ubuntu@ip-172-31-26-5:~$
```

Step VIII : then clone the project repo using “git clone <url_of_repo>

```
ubuntu@ip-172-31-26-5:~$ git clone https://github.com/AgurewJagdale/AngularPipe_Spring-App.git
Cloning into 'AngularPipe_Spring-App'...
remote: Enumerating objects: 173, done.
remote: Counting objects: 100% (173/173), done.
remote: Compressing objects: 100% (159/159), done.
remote: Total 173 (delta 61), reused 0 (delta 0), pack-reused 0
Receiving objects: 100% (173/173), 246.29 KiB | 3.57 MiB/s, done.
Resolving deltas: 100% (61/61), done.
```

Step IX :After that change the directory and append the sql file in database using below command.

```
ubuntu@ip-172-31-26-5:~$ cd AngularSpire Spring-App/
ubuntu@ip-172-31-26-5:~/AngularSpire Spring-App$ ls
README.md  angular-frontend  spring-backend  springbackend.sql
ubuntu@ip-172-31-26-5:~/AngularSpire Spring-App$ mysql -h database-1.crae8ksw8f6q.us-east-2.rds.amazonaws.com -u admin -padmin123 springbackend < springbackend.sql
ubuntu@ip-172-31-26-5:~/AngularSpire Spring-App$
```

Step X : then go to database and check the database and tables in it using below commands.

```

# Import the Keras library, and load the 'mnist' dataset in 'x'
import keras
from keras.datasets import mnist

# Split the data into training and testing sets
(x_train, y_train), (x_test, y_test) = mnist.load_data()

# Convert the data to float32 and normalize it to the range [0, 1]
x_train = x_train.astype('float32') / 255
x_test = x_test.astype('float32') / 255
y_train = y_train.astype('int32')
y_test = y_test.astype('int32')

# Create the model
model = keras.Sequential([
    keras.layers.Flatten(input_shape=(28, 28)),
    keras.layers.Dense(1000, activation='relu'),
    keras.layers.Dense(10, activation='softmax')
])

# Compile the model
model.compile(optimizer='adam',
              loss='categorical_crossentropy',
              metrics=['accuracy'])

# Train the model
model.fit(x_train, y_train, epochs=10,
          validation_data=(x_test, y_test))

# Evaluate the model
test_loss, test_acc = model.evaluate(x_test, y_test)
print('Test accuracy:', test_acc)

```

Step XI : using below command check the table .

```
MySQL [springbackend]> select * from tbl_workers;
```

id	status	workerfname	workerlname
1	Working	Ivan	Holicek
37	Vacation	Marko	Markovic
40	Working	Ivo	Izica
41	Working	Luka	Lukovic
42	Working	Filip	Filipovic

```
5 rows in set (0.001 sec)
```

Step XII: then exit.

Step XIII : Using below command install openjdk-8 .

```

yash@springbackend:~$ exit
yash
hunts@8ip-172-31-26-5:~/Angular$ pip3 install Flask-AppBuilder
Building dependency tree... Done
Building state information... Done
The following additional packages will be installed:
  _adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-common at-spi2-core ca-certificates

```

Step XIV : then install maven using below command.

```
ubuntu@ip-172-31-26-5:~/AngularSpire_Spring-App$ sudo apt install maven
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

Step XV : for establishing database connection with RDS db go to application.properties file and make changes in it

which are given below. Give their rds db endpoint ,username and password of rds.

```
ubuntu@ip-172-31-26-5:~/AngularSpire_Spring-App$ vim spring-backend/src/main/resources/application.properties
```

```
spring.datasource.url=jdbc:mysql://database-1.crae8ksw8f6q.us-east-2.rds.amazonaws.com:3306/springbackend?v=
spring.datasource.username=admin
spring.datasource.password=admin123
spring.jpa.generate-ddl=true
```

Step XVI : then save it.

Step XVII : then build the project using maven .

```
[ERROR] [http-1] http://maven.apache.org/m2whe/dependency/display/MavenMissingProjectException
ubuntu@ip-172-31-26-5:~/AngularSpire_Spring-App$ ls
README.md angular-frontend spring-backend springbackend.sql
ubuntu@ip-172-31-26-5:~/AngularSpire_Spring-App$ cd spring-backend/
ubuntu@ip-172-31-26-5:~/AngularSpire_Spring-App/spring-backend$ mvn clean package -Dmaven.test.skip=true
[INFO] Scanning for projects...
```

Step XVIII : Application is successfully build.

```
Downloaded from central: https://repo.maven.apache.org/maven2/com/google/guava/guava
[INFO] Replacing main artifact with repackaged archive
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 24.566 s
[INFO] Finished at: 2024-06-09T10:32:56Z
[INFO] -----
```

Step XIX : See here, Compiled java application successfully created.

```
ubuntu@ip-172-31-26-5:~/AngularSpire_Spring-App/spring-backend$ ls target/
classes generated-sources maven-archiver maven-status spring-backend-v1.jar spring-backend-v1.jar.original
ubuntu@ip-172-31-26-5:~/AngularSpire_Spring-App/spring-backend$
```

Step XX : Now, run the application using below command “java -jar target/spring-backend-v1.jar”.
Application is run Successfully.

```
classes generated-sources maven-archiver maven-status spring-backend-v1.jar spring-backend-v1.jar.original
ubuntu@ip-172-31-26-5:~/AngularSpire_Spring-App/spring-backend$ java -jar target/spring-backend-v1.jar

Spring
:: Spring Boot ::
(v2.7.4)

2024-06-09 10:35:58.894 INFO 5160 --- [ main] c.e.s.SpringBackendApplication : Starting SpringBackendApplication v1 using Java 1
rSpire_Spring-App/spring-backend/target/spring-backend-v1.jar started by ubuntu in /home/ubuntu/AngularSpire_Spring-App/spring-backend)
```

Step XXI : Now Create the Ec2 Instance for Frontend using ubuntu AMI and connect it.

Successfully terminated i-055f9559-4c07ad1

Instances (2) Info								Refresh	Connect
Find Instance by attribute or tag (case-sensitive)								Running	
☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability		
☐	Backend	i-06af2a77d935bfc3	Running	t2.micro	2/2 checks passed	View alarms +	us-east-2b		
☐	Frontend	i-09dd24fa23ea0f8d1	Running	t2.medium	2/2 checks passed	View alarms +	us-east-2b		

Step XXII : then update it using below packages.

```
ubuntu@ip-172-31-27-85:~$ sudo apt update -y
Get:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:2 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
```

Step XXIII : then install node.js and npm

Step XXIV : then, check the version using “node -v” and “npm -v”.

Step XXV : then install angular cli globally using below command.

```
9.2.0
ubuntu@ip-172-31-27-85:~$ sudo npm install -g @angular/cli@14.2.1
```

Step XXVI : then verify version use command “ng version”

Step XXVII : then clone your git repo .

```
ubuntu@ip-172-31-27-85:~$ git clone https://github.com/ApurvaJagdale/AngularSpire_Spring-App.git
```

Step XXVIII : After that go to your directory.

```
README.md angular-frontend spring-backend springbackend.sql
ubuntu@ip-172-31-27-85:~/AngularSpire_Spring-App$ cd angular-frontend/
ubuntu@ip-172-31-27-85:~/AngularSpire_Spring-App/angular-frontend$
```

Step XXIX: and to connect with backend server go to “vim src/app/services/worker.service.ts”.

Step XXX: and add the public ip of backend server there.

```
@Injectable({
  providedIn: 'root'
})
export class WorkerService {

  private getUrl: string = "http://18.224.165.65:8080/api/v1/workers";

  constructor(private _httpClient: HttpClient) { }

  getWorkers(): Observable<Worker[]> {
    return this._httpClient.get<Worker[]>(this.getUrl).pipe(
      map(response => response)
    )
  }
}
```

Step XXXI : then save it.

Step XXXII : now build it using “ng build” command.

```
ubuntu@ip-172-31-27-85:~/AngularSpire_Spring-App/angular-frontend$ ng build
```

Step XXXIII : See here after building process is completed new directory automatically created.

```
ubuntu@ip-172-31-27-85:~/AngularSpire_Spring-App/angular-frontend$ ls
README.md  angular.json  dist  karma.conf.js  node_modules  package-lock.json  p
ubuntu@ip-172-31-27-85:~/AngularSpire_Spring-App/angular-frontend$
```

Step XXXIV : to start the server use “sudo ng serve --host 0.0.0.0 --port=80” this command.

```
ubuntu@ip-172-31-27-85:~/AngularSpire_Spring-App/angular-frontend$ sudo ng serve --host 0.0.0.0 --port=80
? Would you like to enable autocompletion? This will set up your terminal so pressing TAB while typing Angular
will modify configuration files in your home directory.) Yes
Appended `source <(ng completion script)>` to `/root/.bashrc`. Restart your terminal or run the following to
```

Step XXXV : See here, it compiled Successfully.

```

** Angular Live Development Server is listening on 0.0.0.0:80, open your browser on http://localhost:80/ **

✓ Compiled successfully.
✓ Browser application bundle generation complete.

5 unchanged chunks

Build at: 2024-06-09T11:06:50.763Z - Hash: 6870cbb912ff2c8e - Time: 592ms

✓ Compiled successfully.

```

Step XXXVI : Now hit the public IP of Frontend Server.

← → ↻ ⚠ Not secure 3.23.129.229/workers ☆ Incognito (3) ⋮

Workers

[Add Worker](#)

Order	First Name	Last Name	Status	Edit Button	Delete Button
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Step XXXVII : Now click on Add Worker and Add Workers in it.

← → ↻ ⚠ Not secure 3.23.129.229/workers ☆ Incognito (3) ⋮

Workers

[Add Worker](#)

Order	First Name	Last Name	Status	Edit Button	Delete Button
1.	Apurva	Jagdale	DevOps Intern	Edit	<button>Delete</button>
2.	Rushikesh	Jagdale	FullStack Developer	Edit	<button>Delete</button>
3.	Raavi	Jagdale	aaaa	Edit	<button>Delete</button>

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